

# Detecting Responses to Political Signaling in Social Media

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## Abstract

Implicit political signaling in social media can generate in-group/out-group behavior that affects the way people engage with each other even in apolitical contexts. This study builds up the Twitter roBERTa Sentiment Base model to detect these changes in the way people reply to Reddit posters who signal their political identity through their display name.

## Introduction

Social media has connected the world and enabled ordinary people to interface with orders of magnitude more people on a daily basis than was possible only a few decades ago. This has come with a number of consequences both positive and negative. On one hand, communities of interest can form on incredibly niche topics, sometimes with only a few hundred or thousand members globally, and people within those communities can learn from and help each other. On the other hand, in-group/out-group behavior is intensified by the ability to feed off of like-minded individuals and find reasons to hate the opposing group. Never has this been more true than in the current U.S. political climate, where Donald Trump has leveraged social media to galvanize his political base and launch attacks at his opposition.

The intersection of these two opposing effects provides a highly informative area of study for understanding how people engage with each other. The social media site Reddit is one of the most popular hosts for these shared interest communities, referred to as subreddits, and also often serves as a battleground for politics. Each subreddit has its own culture, many of which are, in theory, not subject to partisanship. In most cases, two people interacting on those apolitical subreddits would be completely unaware of the other's political allegiance. There is one key exception: some users opt to use a display name which signals their politics. In this study, the author asks whether the language people use in responding to a user with a political signal in their display name differs in either subtle or explicit ways. This is accomplished by attempting to learn the difference between responses to these signals using state-of-the-art sentiment analysis techniques. If a machine learning model is able to successfully distinguish between a reply to a liberal signaling username, conservative signaling username, and non-political username using

only the text in the reply, then there must be some underlying change in the way people respond to those usernames.

## Background

Numerous studies have been conducted using Reddit as a source of data. For this study, the Pushshift Reddit Dataset was utilized. In their 2020 paper, dataset creators Baumgartner et. al. detail the contents of the dataset and the 100+ peer-reviewed articles which utilize it. Those articles include topics such as “measuring toxicity, personality, virality, and governance”.<sup>1</sup>

No existing research attempts to build a model that performs the specific classification task in this study: given a response to a social media post, determine whether the post was made by someone with a liberal, conservative, or neutral display name. The closest equivalent is in research by Alkiek et. al., who built two models of interest as part of a larger study on Reddit’s political discourse. The first is an LSTM model to classify usernames as belonging to a user who is liberal vs. conservative. The second is a fine-tuned RoBERTa model to classify text (posts and comments) as written by a user who is liberal vs. conservative. The researchers use multiple methods for determining the political affiliation labels, looking across the entirety of a user’s Reddit activity. Critically, this means there is often no explicit signal in the username being fed to the model, and for the text model they actively removed any examples where the user outright declared their affiliation. The results were mixed: while one version of the text model saw an AUC of 60.43 on their full dataset, all username models performed worse than random.<sup>2</sup>

## Methods

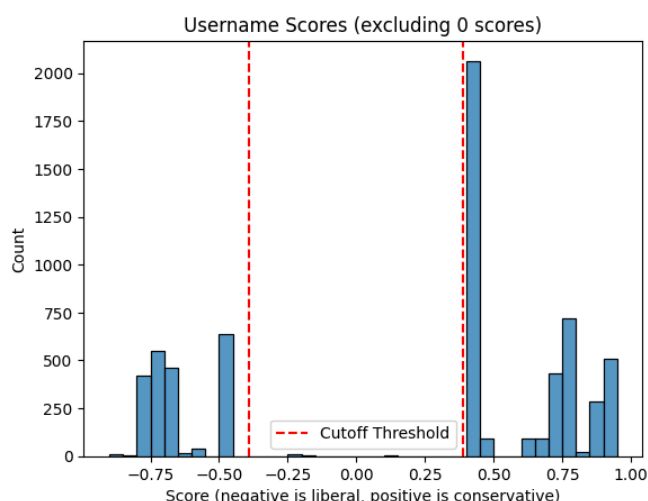
### Data

Direct replies to posts from a total of 41 subreddits, chosen because they appeared by manual review to be apolitical interest-based subreddits, were utilized from the Pushshift Reddit Dataset for this study. These replies were associated with the display name not of the user who wrote the reply but instead of the user who was being replied to.

In order to train the model, the display names needed to be classified as either liberal, conservative, or neutral. While Alkiek et. al. were unable to build a model that could detect underlying political affiliation in usernames, this study only requires identifying the subsets of usernames that, to a social media user quickly glancing, either presents a clear political signal or is clearly apolitical. This was achieved by leveraging the general knowledge of a top-tier LLM at the time of writing, Claude Opus 4.6. Claude was instructed to build a python script which used

regexes to score political signals. It then ran through multiple cycles of executing the script and using the results to refine its methodology.

Out of a total ~5.6 million usernames in the data, 99.9% did not have any political signal detected by the script, 2,138 had a clearly liberal username, and 2248 had a clearly conservative username. This mapped to a total of 12971 replies with a liberal label and 17571 replies with a conservative label. The scoring script did not perform perfectly, for example one of the highest conservative scoring usernames is a false positive where “magazine” was misinterpreted as “maga”, but overall performance was good enough to attempt training a model.



| Highest Scoring Liberal Display Names |       | Highest Scoring Conservative Display Names |       |
|---------------------------------------|-------|--------------------------------------------|-------|
| Display Name                          | Score | Display Name                               | Score |
| FeelTheBernCallTheDr                  | -0.9  | magazinelady                               | 0.95  |
| Canyoufeelthebern                     | -0.9  | MAGA_NW                                    | 0.95  |
| FeelTheBernanke                       | -0.9  | Magaa                                      | 0.95  |
| IFeelTheBern                          | -0.9  | Maga-Munchkin                              | 0.95  |
| feeltheberncream                      | -0.9  | MAGA_ME                                    | 0.95  |
| feelthebern2016                       | -0.9  | magadson                                   | 0.95  |
| FEEL_THE_BERN89                       | -0.9  | Magalicious001                             | 0.95  |
| FeelTheBernVictim                     | -0.9  | magarraaj                                  | 0.95  |
| IFeelTheBernWhenIPee                  | -0.9  | MAGA_AllOverYourAss                        | 0.95  |
| muellersanders2020                    | -0.85 | MagAffect                                  | 0.95  |

All of the replies with a liberal or conservative label, along with an equal number of randomly selected replies with a neutral label, were split 80/20 into train and validation sets. Final evaluation occurred on a set of 2,734 replies from an experiment run by a group including the author where apolitical posts on apolitical subreddits were randomly assigned to either a liberal signaling, conservative signaling, or neutral display name. This data was collected when each post was exactly 24 hours old using web scraping. Further details on this experiment and its application to this study are discussed in the results section.

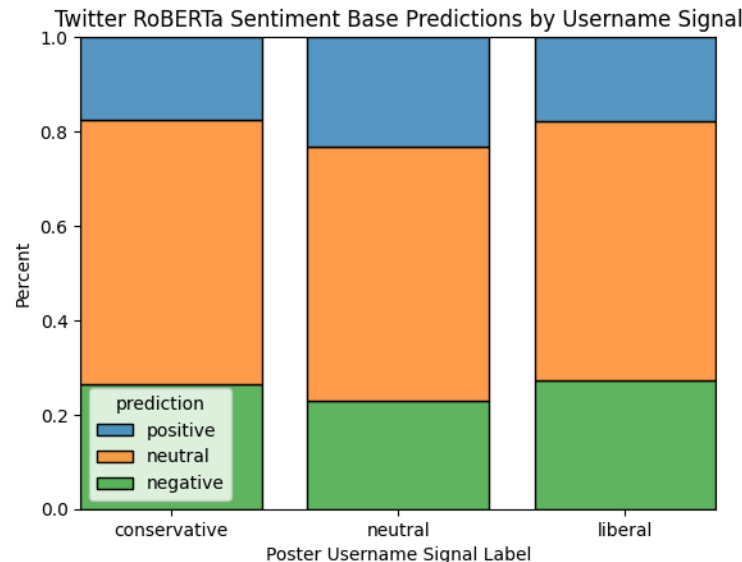
## Model

For this study, a model with strong sentiment analysis performance on social media data was needed as a baseline, so it could be fine tuned on this specific classification task. The model chosen was Twitter RoBERTa Sentiment Base Latest<sup>3</sup>, which holds the highest score on the

TweetEval benchmark<sup>4</sup>. Fine tuning was tested with a number of different layers unfrozen and batch sizes.

## Results

To start, by running an unmodified version of the sentiment analysis model on the data, we can observe that both forms of political signaling username have a higher proportion of negative replies and a lower proportion of positive replies than neutral usernames.



Because the most likely prediction for every label is still neutral, this is not useful for predicting the label. Instead, a simple baseline can be set by predicting neutral for every reply, since the neutral label is roughly twice as common as the other labels in the data.

Running fine tuning across a variety of settings, we see very consistent results with a Macro F1 score of approximately 0.47 on the validation set. This indicates that, at least to the extent possible with the roBERTa architecture, the model is learning all of the information contained in the replies which indicate a reaction to a username's politicization.

| Model                     | Batch Size | Epochs Run | Layers Trained                    | Macro F1 |
|---------------------------|------------|------------|-----------------------------------|----------|
| Always Neutral (Baseline) | -          | -          | -                                 | 0.222    |
| Twitter RoBERTa Sentiment | 32         | 3          | Classifier, Embeddings            | 0.474    |
| Twitter RoBERTa Sentiment | 32         | 3          | Classifier, layer.11, Embeddings  | 0.473    |
| Twitter RoBERTa Sentiment | 32         | 3          | Classifier, Attention, Embeddings | 0.465    |
| Twitter RoBERTa Sentiment | 32         | 3          | All                               | 0.478    |

| Model                            | Batch Size | Epochs Run | Layers Trained                   | Macro F1     |
|----------------------------------|------------|------------|----------------------------------|--------------|
| <b>Twitter RoBERTA Sentiment</b> | <b>16</b>  | <b>5</b>   | <b>All</b>                       | <b>0.479</b> |
| Twitter RoBERTA Sentiment        | 8          | 8          | Classifier, layer.11, Embeddings | 0.474        |

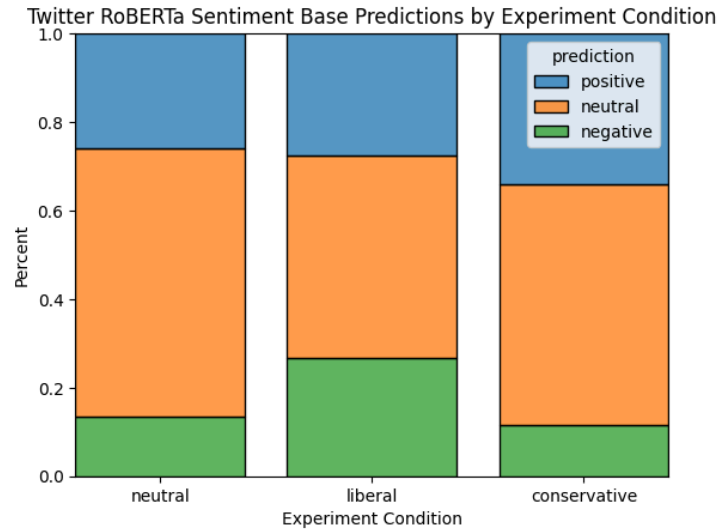
Table 1: Oblation table for various fine tuning settings compared to a baseline on Macro F1

Looking further at the best performing model, it performs best on neutral posts in both precision and recall. One possible explanation for this is that reactions to liberal and conservative signaling posts have some similarities, for example a conservative replying to a liberal-signaler could exhibit some of the same out-group behaviors as a liberal replying to a conservative-signaler. Between the two signaling labels, precision is very similar, but recall is much higher for conservative. This likely means that a higher proportion of replies to conservative posters have information which indicates a response to the signal when compared to liberal posters.

| <b>Best Performing</b> | Liberal | Conservative | Neutral |
|------------------------|---------|--------------|---------|
| Precision              | 0.456   | 0.452        | 0.585   |
| Recall                 | 0.292   | 0.470        | 0.662   |
| F1                     | 0.356   | 0.461        | 0.621   |

Table 2: In-depth metrics for the best performing model on the validation set

One issue with looking at observational data from historical reddit posts is that there may be a confounding variable which is correlated with both a user’s display name and other characteristics about their post. For example, someone with a conservative signaling username is almost certainly conservative, and so might be more likely to post on different subreddits with different types of posts or with otherwise different behaviors. To account for this, a group including the author ran an experiment where everything from subreddit to post content to post date was randomized, removing the possibility of any confounders. We see when running both sentiment analysis and the best fine tuned model on this data that, there is a noticeable increase in negative replies to liberal posts and the best model is somewhat more reliable than the always-neutral predictor but performs far worse than on validation data. This likely means that the model was partially detecting a response to the username and partially detecting some other confounding characteristic.



| F1         | Liberal | Conservative | Neutral | Macro |
|------------|---------|--------------|---------|-------|
| Best Model | 0.150   | 0.238        | 0.457   | 0.282 |
| Baseline   | 0       | 0            | 0.500   | 0.167 |

Table 3: Best performing model vs. a baseline that always predicts neutral

| Best Performing | Liberal | Conservative | Neutral |
|-----------------|---------|--------------|---------|
| Precision       | 0.307   | 0.253        | 0.354   |
| Recall          | 0.099   | 0.226        | 0.644   |
| F1              | 0.150   | 0.238        | 0.457   |

Table 4: In-depth metrics for the best performing model on the experiment data

Examples of the top 5 ranked replies to both political signal groups are included in the appendix.

## Conclusion

This study demonstrates that there is a measurable difference in replies a Reddit poster who signals their political identity will see compared to one who doesn't. It quantifies this difference by using machine learning and natural language processing to detect the inherent differences in these replies, leveraging posts from a randomized experiment to ensure that the results are not due to confounding variables that affect both the username of a poster and the types of posts they make.

# References

- <sup>1</sup> Baumgartner, Jason, et al. “The pushshift reddit dataset.” *Proceedings of the International AAAI Conference on Web and Social Media*, vol. 14, 26 May 2020, pp. 830–839, <https://doi.org/10.1609/icwsm.v14i1.7347>.
- <sup>2</sup> Alkiek, K., Zhang, B., & Jurgens, D. (2022). Classification without (Proper) Representation: Political Heterogeneity in Social Media and Its Implications for Classification and Behavioral Analysis. *Findings of the Association for Computational Linguistics: ACL 2022*, 504–522. <https://doi.org/10.18653/v1/2022.findings-acl.43>
- <sup>3</sup> Loureiro, Daniel, et al. “TimeLMs: Diachronic language models from Twitter.” *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics: System Demonstrations, 2022*, pp. 251–260, <https://doi.org/10.18653/v1/2022.acl-demo.25>.
- <sup>4</sup> Barbieri, Francesco, et al. “Tweeteval: Unified benchmark and comparative evaluation for Tweet Classification.” *Findings of the Association for Computational Linguistics: EMNLP 2020, 2020*, pp. 1644–1650, <https://doi.org/10.18653/v1/2020.findings-emnlp.148>.

## Appendix: Examples

The below table has examples of the highest scoring liberal and conservative predictions from the best model:

| Most Conservative                                                                                                                                                                                                                                                                                                            | Most Liberal                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Not me....people who go to quick oil change places.                                                                                                                                                                                                                                                                          | The unsubscribe button                                                                                                                                                                                                                                                                                                                                                                                                           |
| There are enough bozos that drive aggressively that I see one every time I'm on the highway but most people are moving with traffic, signaling, and also frustrated by the bozos.                                                                                                                                            | I AM Not Okay With This. Was one of the COVID shows that was renewed and then cancelled.                                                                                                                                                                                                                                                                                                                                         |
| Did my car brakes myself. Started smoking on the way to work. I lost a days pay and had to go to the mechanic to get them fixed.\nI will still do them again but next I'll watch for seized calipers                                                                                                                         | Actually looking through the viewfinder. As in looking around the edges of the shot and at the (not the subject) items in the shot in order to decide if I needed to reframe. Thinking about what should or should not be in the image before pressing the shutter button                                                                                                                                                        |
| When it's a communal public spaces, everyone is relatively quiet and keeps their voices low and is respectful, and then that one group comes in, loud and chattering like they own the space, standing around for an hour or so loud enough that you can hear them down the hall like there aren't people trying to do work. | I've gotten really into Find Me in Paris, the only annoying thing is Netflix only has season 1. Since I have Hulu and Disney+ I'm able to watch the other 2 seasons though so it's ok.                                                                                                                                                                                                                                           |
| Their muffins are insane, my non celiac gf constantly steals them                                                                                                                                                                                                                                                            | Sense 8, The O A, Kaos... but they cancel most all the good ones. Quit Netflix for almost a year, signed back up to watch One Piece s2 (insanely good btw), but that was literally the only new thing after 9 months that seemed interesting or worthwhile. Netflix has crapped out, they'll have 1 or 2 good shows all year, and then cancel half of them. I hate getting too into a Netflix show anymore for that reason, too. |